THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The slope of the line $2x + 2y = 10$ is: 1

(a) -2 **(b)** 1 **(c)** -1 **(d)** 0

2. The points $(3, 6)$, $(5, 10)$ and $(7, k)$ are collinear. The value of k is: 1

(a) 14 **(b)** 15 **(c)** 16 **(d)** 13

3. The line $4x + 2y = 9$ is parallel to the line $kx + 4y = 11$. The value of k is: 1

(a) 8 **(b)** 10 **(c)** 9 **(d)** 7

4. The line $3x + 2y = 6$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

5. A taxi fare is ₹83 for 5 km and ₹193 for 15 km (fare is a linear function of distance). The fare for 21 km (in ₹) is: 1

(a) 248 **(b)** 259 **(c)** 270 **(d)** 287

6. The point $(-4, k)$ lies on the line $2x + 6y = 28$. The value of k is: 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

7. The pair of equations $4x + 5y = 4$ and $4x + 7y = 7$ has: 1

(a) exactly one solution **(b)** exactly two solutions

(c) infinitely many solutions **(d)** no solution

8. A line with slope 6 passes through $(0, 7)$. Its y-intercept is: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

9. The slope of the line $3x + 6y = 15$ is: 1

(a) $-\frac{3}{2}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{1}{2}$ **(d)** $-\frac{1}{2}$

10. The points $(-2, -1)$, $(-1, -4)$ and $(0, k)$ are collinear. The value of k is: 1

(a) -8 **(b)** -6 **(c)** -7 **(d)** -10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A taxi fare is ₹82 for 3 km and ₹163 for 12 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 23 km. 2

12. Find k if the lines $3x + 5y = 9$ and $kx + 10y = 11$ are parallel. Why are they not the same line? 2

13. Write $6x - 2y = 18$ in the form $y = mx + c$ and read off the slope and the y-intercept. 2

14. A taxi fare is ₹140 for 6 km and ₹259 for 13 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 28 km. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the triangle formed by the line $8x + 4y = 32$ and the coordinate axes. 3

16. Find k if the points $(-3, 5)$, $(-2, 2)$ and $(-1, k)$ lie on one straight line. 3

17. Find the equation of the line through $(-1, -5)$ and $(-6, -25)$ in the form $y = mx + c$. 3

OR

Find the area of the triangle formed by the line $2x + 9y = 18$ and the coordinate axes. 3

18. Find k if the points $(-9, 0)$, $(-7, -4)$ and $(-5, k)$ lie on one straight line. 3

OR

Find the equation of the line through $(-4, 2)$ and $(5, -7)$ in the form $y = mx + c$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A printing shop charges ₹50 for 20 pages and ₹90 for 40 pages (linear charge). Find the equation of the charge y in terms of the number of pages x, the charge for 128 pages and the number of pages for which the charge is ₹286. 5

OR

Without solving, state how many solutions the pairs $4x + y = 7,\ 8x + 2y = 14$ and $4x + y = 7,\ 8x + 2y = 16$ have. 5

20. A printing shop charges ₹107 for 10 pages and ₹638 for 69 pages (linear charge). Find the equation of the charge y in terms of the number of pages x, the charge for 167 pages and the number of pages for which the charge is ₹1610. 5

OR

Without solving, state how many solutions the pairs $2x + 5y = 8,\ 8x + 20y = 32$ and $2x + 5y = 8,\ 8x + 20y = 36$ have. 5

END OF QUESTION PAPER — **Total: 40 Marks**